

MALLMART

On-Demand Mall Shopping & Delivery Platform

Project Documentation

Prepared for Iconic — Ghana

Version 1.0 · July 2026

Customer → Mall Shopper → Dispatch Rider → Delivered

Overseen end-to-end by a Supervisor and the Owner/Admin dashboard

Contents

1. Overview

MallMart is an on-demand shopping and delivery platform built for the Ghanaian market. A customer shops from home for items sold inside a physical mall; a mall-based staff member (the “Shopper”) buys those items in person on the customer's behalf and pays the store directly by Mobile Money; a dispatch rider then collects the finished order and delivers it to the customer's address. A supervisor watches every order move through this chain in real time, and the business owner has a separate dashboard covering revenue and operations across the whole platform.

The defining rule of the model: the customer pays the full amount — items, delivery fee, and service charge — by Mobile Money before any shopping begins. That single payment is the same pool of funds the Shopper later uses to pay the mall store, so nothing is initiated on trust; the money is already in hand before a Shopper is asked to spend anything on the customer's behalf.

This document describes the five interfaces, the order lifecycle that connects them, the data and fee model, what has been built as a clickable prototype so far, and what is still needed to take MallMart into production.

2. The Five Interfaces

MallMart is really one shared order, viewed and acted on differently by five kinds of user. Each interface only shows the actions relevant to that person's job.

2.1 Customer App

- Browse malls, then stores inside a chosen mall, then products inside a chosen store.
- Build a cart across one or more stores in the same mall.
- Enter delivery details — name, phone number, and delivery location.
- Pay the full order total up front via MTN MoMo, Vodafone Cash, or AirtelTigo Money.
- Track the order on a live status timeline from payment through to delivery, including which Shopper and which Rider are handling it.

2.2 Mall Shopper App

- See a queue of newly paid orders assigned to their mall and accept one to start shopping.
- Work through an item checklist inside the store, ticking off each item as it is picked.
- Once every item is picked, confirm the Mobile Money payment made to the store for the exact subtotal.
- Hand the finished, paid-for order to a dispatch rider to close out their part of the job.

2.3 Dispatch Rider App

- See a list of orders that are packed and waiting for pickup at the mall counter.
- Accept a delivery, which reveals the customer's name, phone number, delivery location, and distance from the mall.
- Mark the order delivered once it reaches the customer, closing the loop.

2.4 Supervisor Console

- A live table of every order in the system — who is shopping, who is riding, and what stage each order is at.

- Filter by active or flagged orders, and search by order ID, customer, or mall.
- Flag an order that needs attention (a delayed rider, a customer complaint, a payment mismatch) and attach a note for the record.

2.5 Owner / Admin Dashboard

- Gross order value and delivery-plus-service revenue at a glance.
- Active order and flagged order counts.
- A revenue trend chart and a full ledger of every order with its Mobile Money provider and total.
- Visibility over how many shoppers and riders are on duty.

3. Order Lifecycle

Every order in MallMart moves through the same seven stages, in the same order, regardless of which mall or store it came from. This is the backbone that all five interfaces read from and write to.

#	Stage	What happens	Who acts
1	Payment Confirmed	Customer pays the full total (items + delivery + service charge) by Mobile Money. Nothing else starts until this clears.	Customer
2	Shopper Assigned & Shopping	A Shopper at the mall accepts the order and physically picks each item on the checklist.	Mall Shopper
3	Paid at Mall	Once every item is picked, the Shopper pays the store the exact subtotal by Mobile Money, using funds from the customer's original payment.	Mall Shopper
4	Waiting for Rider	The paid, packed order is handed to the dispatch counter and becomes visible to available riders.	Mall Shopper → Dispatch Rider
5	Out for Delivery	A rider accepts the order and travels to the customer's location.	Dispatch Rider
6	Delivered	The rider confirms hand-off to the customer. The order is closed.	Dispatch Rider

Table 1 — Order stages, in the order every order passes through them. Supervisor and Admin can see this stage for any order at any time; only the Customer's own order is visible in the Customer app.

4. Payment Model

4.1 Why payment happens first

Because the Shopper is spending the company's (and effectively the customer's) money inside the mall, MallMart never sends a Shopper to buy anything before the customer's payment has cleared. This protects the business from fronting cash on unconfirmed orders and protects the Shopper from being asked to pay out of pocket.

4.2 What the customer pays

Component	How it's calculated	Example (GH¢232 cart)
Items subtotal	Sum of item price × quantity, at the price shown in the app	GH¢232.00

Component	How it's calculated	Example (GH¢232 cart)
Delivery fee	Flat fee per order	GH¢25.00
Service charge	5% of the items subtotal, GH¢10 minimum	GH¢12.00
Total charged to customer	Items + delivery + service charge	GH¢269.00

Table 2 — Fee structure as implemented in the prototype. The delivery fee and service-charge percentage are both easy to make configurable per mall, per distance band, or per order size in production.

4.3 What the Shopper pays

The Shopper pays the mall store only the items subtotal — the exact amount shown on the in-app checklist — by Mobile Money at the till. The delivery fee and service charge stay with the company; they are never handed to the store.

5. Data Model

Everything in the prototype revolves around a single Order record. In production this becomes the central table in the platform's database, with related tables for users, malls, stores, and products.

Field	Type	Description
id	String	Unique order reference, e.g. ORD-1006
status	Enum	One of the seven lifecycle stages in Table 1
customerName / Phone / Location	String	Delivery contact details captured at checkout
mallName / storeName	String	Where the order is being shopped
items[]	Array	Name, price, quantity, and a picked flag per line item
subtotal / deliveryFee / serviceFee / total	Number	Fee breakdown as in Table 2
momoProvider	Enum	MTN MoMo, Vodafone Cash, or AirtelTigo Money
shopper / rider	String	Staff member currently assigned, once assigned
flagged / flagNote	Boolean / String	Set by a Supervisor when an order needs attention

Table 3 — Core order fields used across all five interfaces.

6. Current Prototype

What exists today is a clickable, front-end-only prototype (React) that demonstrates the full loop with realistic mock data — six seed orders sitting at different stages, three mock stores across three real Ghanaian malls, and a role switcher so any interface can be previewed instantly.

6.1 What it proves

- The five interfaces and the information each role needs to do their job.
- The order lifecycle and how a single order visibly moves from one interface to the next.

- The fee model and Mobile Money checkout flow, end to end.
- The look and feel of a live operations view (Supervisor) and a business dashboard (Admin).

6.2 What it does not yet do

- No real backend, database, or persistence — refreshing the page resets all data.
- No real Mobile Money integration — payment is simulated by a button press.
- No authentication — anyone can open any of the five interfaces.
- No real-time sync between devices — in production, a Shopper accepting an order needs to instantly disappear from every other Shopper's queue.
- No maps — the rider's map is a placeholder; production needs real geolocation and routing.

7. Recommended Production Architecture

To turn this into a live product, the prototype's front-end screens can largely be kept — the work is in wiring them to a real backend. A reasonable shape:

7.1 Backend & data

- A central API (e.g. Node.js/Express or similar) backed by a relational database (PostgreSQL) with tables for users, orders, order_items, malls, stores, products, and payments.
- Role-based authentication (phone number + PIN, or email + password) with four staff roles — Shopper, Rider, Supervisor, Admin — plus Customer accounts.
- A real-time layer (WebSockets or a push service) so that when an order changes stage, every relevant interface updates without a manual refresh — this matters most for the Shopper queue and the Rider's available-pickups list.

7.2 Mobile Money integration

- Direct integration with each provider's collections API (MTN MoMo, Vodafone Cash, AirtelTigo Money), or a Ghanaian payment aggregator that supports all three (e.g. Paystack, Flutterwave, Hubtel) so one integration covers all networks.
- Customer payment should be a collection request that the customer approves on their phone, with a webhook confirming success before the order moves to “Shopper Assigned.”
- The Shopper's payment to the store should ideally also run through the same MoMo integration (a disbursement or a second collection against the held funds), so there is a digital record matching what was picked to what was paid — rather than the Shopper paying manually and self-reporting.
- A reconciliation step for when the actual mall total differs from the app's cart total (price changes, out-of-stock substitutions, promotions) — decide in advance whether the Shopper, Supervisor, or an automatic refund/top-up handles the difference.

7.3 Rider tooling

- Real geolocation for both the rider and the customer's delivery pin, with a maps SDK (Google Maps or Mapbox) for routing and live position sharing.
- Push notifications for new available pickups and for the customer as their order moves.

7.4 Admin & reporting

- Exportable reports (daily settlement, revenue by mall, staff performance) rather than only an in-app dashboard.

- An audit trail on every status change — who changed it and when — which the Supervisor's flag notes in the prototype hint at but don't yet fully log.

8. Roles & Responsibilities Summary

Role	Primary responsibility	Cannot do
Customer	Choose items, pay up front, track delivery	See other customers' orders
Mall Shopper	Pick items in-store, pay the store, hand off to dispatch	Assign riders or edit fees
Dispatch Rider	Collect from the mall, deliver to the customer	See orders not yet packed
Supervisor	Monitor every order, flag and resolve issues	Change fees or view company-wide revenue
Owner / Admin	Full visibility — revenue, operations, staffing	— (full access)

9. Suggested Next Steps

1. Confirm the fee model (flat delivery fee vs. distance-based, service charge percentage) before it's hard-coded into a backend.
2. Choose a Mobile Money integration partner and confirm settlement timing — this affects how quickly a Shopper can be told funds are ready to spend.
3. Define staff onboarding — how a new Shopper or Rider account is created and assigned to a mall or zone.
4. Decide the reconciliation policy for mall-price mismatches, referenced in section 7.2.
5. Move from the current mock-data prototype to a connected backend, starting with the Customer checkout → Shopper queue link, since that is the core trust-critical handoff.

This document describes the MallMart concept and prototype as of July 2026. It will need updating as fee rules, payment integrations, and staff workflows are finalised.